You will build an interactive web application that demonstrates fundamental JavaScript concepts within an HTML page. This project will help you understand DOM manipulation, event handling, user input processing, and basic web development principles.

Required Features

Feature 1: Basic Button Interactions

What to build:

1. Create 3 buttons that perform different actions when clicked
2. Button 1: Show an alert message
3. Button 2: Change the page background color
4. Button 3: Display the current date and time

Technical requirements:

1. Use onclick attributes or event listeners
2. Access elements using getElementById()
3. Modify element content using innerHTML

Feature 2: User Input & Calculator

What to build:

1. A greeting section that takes user's name and displays a personalized message
2. A simple calculator that performs basic math operations (+, -, ×, ÷)
3. Input validation to handle empty or invalid inputs

Technical requirements:

1. Use <input> elements to get user data
2. Validate input before processing
3. Use switch or if-else statements for calculator logic
4. Handle edge cases (like division by zero)

Feature 3: Todo List Application

What to build:

1. Add new tasks to a list
2. Mark tasks as completed (with visual indication)
3. Delete tasks from the list
4. Display message when no tasks exist

Technical requirements:

1. Store todos in an array of objects
2. Each todo should have: id, text, completed status
3. Use array methods like push(), filter(), map()
4. Generate HTML dynamically based on array data

Feature 4: Random Quote Generator

What to build:

1. Display a random inspirational quote when button is clicked
2. Store quotes in an array
3. Show different quote each time

Technical requirements:

1. Create an array of at least 5 quotes
2. Use Math.random() to select random quotes
3. Display quotes in a designated area

Implementation Steps

Step 1: Set up HTML Structure

1. Create basic HTML5 document structure
2. Add a title and meta tags
3. Create sections for each feature using <div> elements
4. Add headings (<h1>, <h2>) for each section
5. Include necessary form elements: buttons, inputs, select dropdown

Step 2: Style with CSS

1. Add internal CSS within <style> tags in the <head>
2. Style the page layout (container, sections, spacing)
3. Design buttons with hover effects
4. Create visual feedback for completed todos
5. Make the page responsive and visually appealing

Step 3: Implement JavaScript Functions

Step 4: Testing & Debugging

1. Test each feature individually
2. Check for errors in browser console (F12)
3. Validate inputs with edge cases
4. Ensure responsive design works on different screen sizes